SKILLBRIDGE

SMART SKILL DEVELOPMENT PLATFORM

MCA Full-Stack Web Application Project Report

Abstract

SkillBridge is a web-based skill development platform designed to help learners discover structured courses, enroll in learning paths, complete lessons, take knowledge assessments and monitor progress. The system uses Python and Django for server-side functionality, HTML/CSS/JavaScript for the responsive interface, and SQLite by default with optional PostgreSQL support.

1. Introduction

Digital learning resources are widely available, but learners may find it difficult to organize learning, monitor progress and evaluate their understanding. SkillBridge brings course discovery, lesson tracking, assessments and a learning dashboard into one application.

2. Problem Statement

Learners often use disconnected tools for course discovery, study materials, progress tracking and self-assessment. This makes it harder to maintain a consistent learning path and review progress.

3. Objectives

Provide student authentication and profiles; publish and search courses; allow course enrollment; track lesson completion; evaluate knowledge through quizzes; summarize progress; and present goal-oriented course suggestions.

4. Scope

The academic implementation covers student accounts, course and lesson management, enrollment, lesson completion, quizzes, automatic scoring, a dashboard, basic recommendations and administrator management. Payments, live classes, proctoring and production certificate signing are outside the current scope.

5. Technology Stack

Backend: Python and Django. Frontend: HTML5, CSS3 and JavaScript. Database: SQLite default; PostgreSQL optional. Development tools: VS Code, Git and web browser.

6. System Modules

Accounts: registration, login, logout and learner profile. Courses: catalogue, search, course details and enrollment. Learning: ordered lessons and completion records. Assessments: multiple-choice questions, automatic scoring and pass status. Progress: dashboard metrics and course recommendations. Admin: manage course, lesson, quiz and question data.

7. System Design

The browser sends HTTP requests to Django URL routes. Views validate requests and apply business rules. Django models communicate with the relational database. Templates render server-side HTML and static CSS/JavaScript provide presentation and lightweight interactions. Authentication is provided by Django's built-in user system.

8. Database Design

Course stores title, slug, category, description, skills, level and duration. Lesson belongs to a course. Enrollment links a learner to a course and prevents duplicate enrollment. LessonCompletion links an enrollment to a completed lesson. Quiz belongs to a course; Question belongs to a quiz; QuizAttempt stores the learner's score, total and pass result. Profile stores learner headline, career goal and bio.

9. Implementation

The project is organized into accounts, courses, assessments and progress Django applications. Administrators create courses, lessons and quizzes in Django admin. Learners register, enroll, complete lessons and submit quiz answers. The dashboard calculates progress from lesson completion records.

10. Testing Strategy

Run Django system checks, migration consistency checks and automated tests. Manual testing should cover registration, login/logout, profile editing, course search, enrollment, lesson completion, dashboard progress, quiz submission, result visibility and admin operations. Add project-specific automated tests before final academic submission.

11. Installation

Create a virtual environment, install requirements, run python manage.py migrate, create an administrator with python manage.py createsuperuser, optionally populate sample content with python manage.py seed_demo, and start with python manage.py runserver. Open http://127.0.0.1:8000/.

12. Limitations and Future Enhancements

Current recommendations use transparent keyword matching rather than a trained AI model. Future work may add AI-based skill-gap analysis, richer analytics, video hosting, downloadable certificates, email verification, notifications, course ratings, APIs and deployment hardening.

Conclusion

SkillBridge demonstrates a modular full-stack learning platform with learner accounts, courses, lesson tracking, assessments and progress summaries. It provides a foundation that can be extended with more advanced personalization and deployment capabilities.

Appendix: Verification Commands

python manage.py check
python manage.py makemigrations --check --dry-run
python manage.py test
python manage.py seed_demo